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Expanding Clinical Spectrum of Anti-GQ1b Antibody Syndrome
A Review

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IMPORTANCE The discovery of the anti-GQ1b antibody has expanded the nosology of classic Miller Fisher syndrome to include Bickerstaff brainstem encephalitis, Guillain-Barré syndrome with ophthalmoplegia, and acute ophthalmoplegia without ataxia, which have been brought under the umbrella term *anti-GQ1b antibody syndrome*. It seems timely to define the phenotypes of anti-GQ1b antibody syndrome for the proper diagnosis of this syndrome with diverse clinical presentations. This review summarizes these syndromes and introduces recently identified subtypes.

OBSERVATIONS Although ophthalmoplegia is a hallmark of anti-GQ1b antibody syndrome, recent studies have identified this antibody in patients with acute vestibular syndrome, optic neuropathy with disc swelling, and acute sensory ataxic neuropathy of atypical presentation. Ophthalmoplegia associated with anti-GQ1b antibody positivity is complete in more than half of the patients but may be monocular or comitant. The prognosis is mostly favorable; however, approximately 14% of patients experience relapse.

CONCLUSIONS AND RELEVANCE Anti-GQ1b antibody syndrome may present diverse neurological manifestations, including ophthalmoplegia, ataxia, areflexia, central or peripheral vestibulopathy, and optic neuropathy. Understanding the wide clinical spectrum may aid in the differentiation and management of immune-mediated neuropathies with multiple presentations.

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Gangliosides are glycosphingolipids containing at least 1 sialic acid linked to the carbohydrate moiety.[1,2] They are believed to play a role in neuroplasticity by influencing neurogenesis, synaptic plasticity, neurotransmission, and axonal growth, thereby enabling neuronal differentiation, maintenance, and repair.[3,4] Antibodies targeting the gangliosides are implicated in many autoimmune peripheral neuropathies.[5] More than 20 antibodies against the gangliosides have been identified in association with neurological disorders,[6] with GM1, GD1b, and GQ1b being the major targets. Although a single antibody usually binds to a specific epitope to give rise to a certain neurological disorder, the antiganglioside antibody can also bind to a ganglioside complex, creating a new epitope, or to other antigens in case of structural similarity.[1,7,8] For instance, anti-GQ1b antibody can bind to complex clusters of glycolipids containing the GQ1b (ie, antiganglioside complex antibody against GQ1b/GM1 or GQ1b/GD1a)[8] or also can cross-react with different individual gangliosides sharing a disialosyl structure (ie, GD1b and GT1a).[9]

Classic Miller Fisher syndrome (MFS) is characterized by the clinical triad of external ophthalmoplegia, ataxia, and areflexia.[10] The diagnosis was based on clinical presentation[10] until the identification of the anti-GQ1b antibody in this disorder.[11] The discovery of this antibody has further expanded the nosology of this syndrome.[12] MFS is a subtype of anti-GQ1b antibody–related neurological syndromes that also include acute ophthalmoplegia without ataxia,[13,14] Bickerstaff brainstem encephalitis (BBE),[15,16] and Guillain-Barré syndrome with ophthalmoplegia (GBS-O).[17]

There are also limited forms of anti-GQ1b antibody syndrome that present with 2 or 1 of the clinical triad of MFS, complicating the diagnosis.[18] The recently identified subtypes, eg, acute vestibular syndrome[19,20] or optic neuropathy,[21,22] lack ophthalmoplegia, which is the most prominent feature in anti-GQ1b antibody syndromes.[19] It seems timely to define the phenotypes of anti-GQ1b antibody syndrome for the proper diagnosis of this syndrome with diverse clinical presentations.

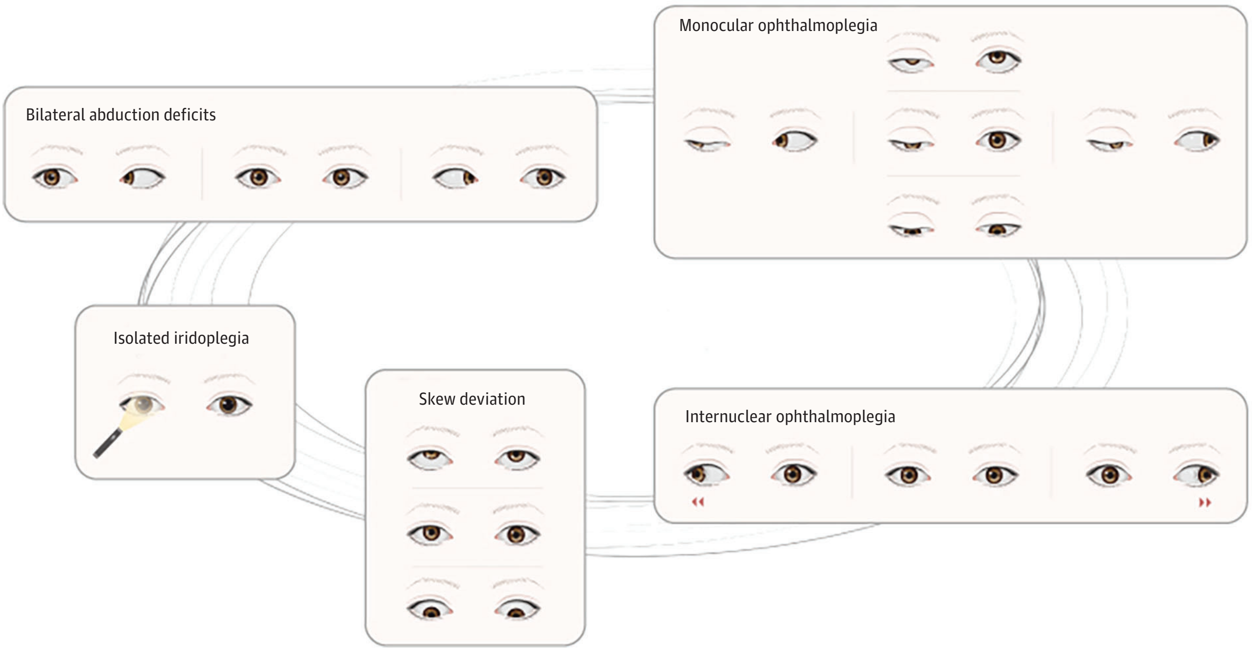
Clinical Features of the Subtypes

Typical Anti-GQ1b Antibody Syndrome With Ophthalmoplegia

Classic MFS

Classic MFS is characterized by the triad of external ophthalmoplegia, ataxia, and areflexia (**Figure 1**).[10] The pathogenetic role of the anti-GQ1b antibody in the development of classic MFS has been recognized since the early 1990s.[11] Classic MFS is the most common phenotype of anti-GQ1b antibody syndrome.[19] The antibody titer increases during the early acute phase and declines with time, whereas the antibody is not detected in healthy controls or disease control patients.[11,17] The pathogenic role of anti-GQ1b antibodies is further

Figure 1. Illustration of the Spectrum of Ophthalmoplegia Observed in Anti-GQ1b Antibody Syndrome



The patterns of ophthalmoplegia can range from isolated iridoplegia to complete external/internal ophthalmoplegia.

supported by their early detection during the disease course, within 1 or 2 days after the onset of neurologic deficits. This implies that the antibody is indeed involved in the pathogenesis and not an epiphenomenon due to tissue destruction.[17] However, this should be interpreted with caution because anti-GQ1b antibody positivity can also be an epiphenomenon in the presence of another autoimmune disorder or dysimmune state. Anti-GQ1b antibodies are also found in other clinical phenotypes, and these disorders are grouped under the umbrella term *anti-GQ1b antibody syndrome*.[10,12]

Overlap With GBS

Ophthalmoplegia can be observed in 15% of GBS cases.[23] Motor weakness is prominent, graded 3 or less on the Medical Research Council scale. If the clinical phenotype is also compatible with other subtypes (eg, BBE or MFS) while manifesting as a prominent appendicular weakness, it is considered to overlap with GBS (ie, GBS-O).[12,24] Patients with GBS-O account for 20% to 28% of those with anti-GQ1b antibody syndrome.[12,19]

BBE

A clinical diagnosis of BBE can be made when patients show mental changes with pyramidal signs and ophthalmoplegia.[11,12,15] The consciousness disturbance ranges from drowsiness (about 43%) to coma.[12,24] Apart from the pathologic reflexes (eg, Babinski sign or deep tendon hyperreflexia), mild appendicular weakness can be associated.[12] If the weakness is profound, an overlap with GBS should be considered.[24] Other than those core features, various focal neurologic deficits and nystagmus can be observed in patients with BBE.[25] Brain magnetic resonance images (MRIs) are usually unrevealing but may be required to exclude potential mimickers, such as brainstem stroke, Wernicke encephalopathy, and infectious, demyelinating, or autoimmune encephalitis.[26] Approximately 10% to 30% of patients with BBE show high signal-intensity lesions in the mesencephalon, cerebellum, or thalamus on T2-weighted MRI (**Figure 2**).[24,26,27] The clinical outcome of BBE is generally favorable unless overlapping with GBS, and 66% of patients show a full recovery within 6 months.[24]

Acute Ophthalmoplegia Without Ataxia

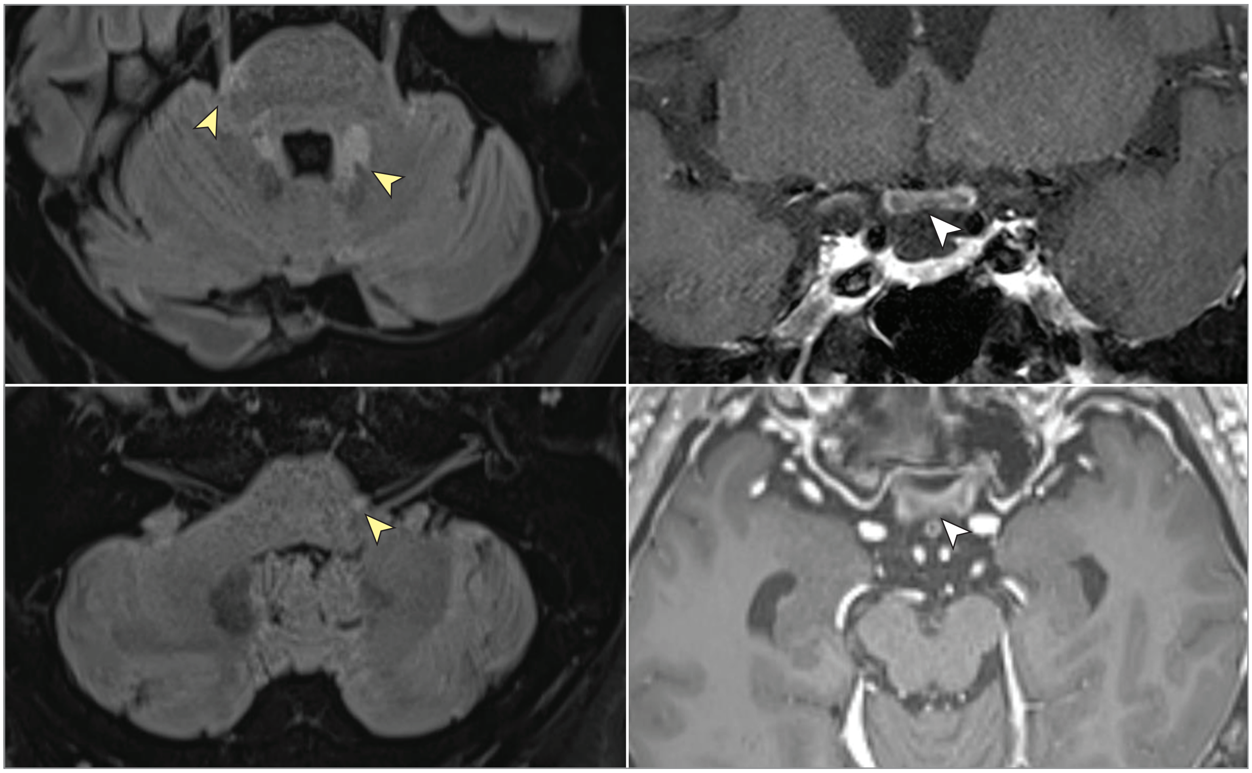
Some patients with anti-GQ1b antibodies may present with a limited form of acute ophthalmoplegia without areflexia or ataxia (Figure 1).[14,28] In this instance, a nerve conduction study and cerebrospinal fluid (CSF) analysis, which would guide the diagnosis of anti-GQ1b antibody syndrome, rarely show abnormal results.[13,14] Thus, ocular motor findings are important because they are the only clinical signs for diagnosis before the results of antibody tests are available. Bilateral involvement is common (73%), mostly with abduction deficits in both eyes.[28] Iridoplegia may be observed in nearly half of the patients.[14] As external ophthalmoplegia may take the feature of multiple ocular motor nerve palsies without other signs of polyradiculopathies, differentiation from myasthenia gravis, thyroid-associated ophthalmopathy, Wernicke encephalopathy, or Tolosa-Hunt syndrome can be challenging.

Atypical Antiganglioside Antibody Syndrome Without Ophthalmoplegia

Acute Vestibular Syndrome

Although GQ1b gangliosides are preferentially expressed in the ocular motor nerves, they are also present in a high concentration in the vestibulocochlear nerves.[29] Therefore, canal paresis can be observed during bithermal caloric tests in patients with MFS but without ophthalmoplegia.[30] Furthermore, typical features of acute uni-

Figure 2. Magnetic Resonance Images of a Patient With Bickerstaff Brainstem Encephalitis



Multiple scattered lesions are found in the brainstem (yellow arrowheads) and optic chiasm (white arrowheads).

lateral peripheral vestibulopathy (ie, vestibular neuritis) may be observed in association with serum anti-GQ1b antibody positivity (**Figure 3**, **Figure 4**, and **Table**).[20,31] Indeed, antiganglioside antibodies, mostly anti-GQ1b antibodies, were found in 12 of 105 patients (11%) with acute unilateral peripheral vestibulopathy.[31] In these cases, abnormal head-impulse test results[32] and canal paresis on bithermal caloric tests[30,33-35] mostly indicate peripheral vestibulopathy. Furthermore, spontaneous downbeat nystagmus; periodic alternating, gaze-evoked, and central positional nystagmus; ocular flutter; and saccadic dysmetria have been described in association with anti-GQ1b antibody positivity.[30,32,36,37] Thus, anti-GQ1b antibody syndrome should be considered in patients presenting with either peripheral or central vestibulopathy of a probable immune-mediated etiology.

Optic Neuropathy With Disc Swelling

Since GQ1b is also abundant in the optic nerve,[29] optic neuropathy can be observed in anti-GQ1b antibody syndrome.[16,38,39] However, there have only been anecdotes of optic nerve involvements in association with anti-GQ1b antibody positivity in the literature.[22,38-41]

Optic nerve involvement may be either unilateral or bilateral,[22] and it can occur during the recovery phase of ophthalmoplegia.[39] Patients almost always show disc swelling, even though the mechanism remains unclear. This may be due to inflammation involving the optic nerves (ie, optic neuritis) or secondary to increased intracranial pressure and stasis of the axoplasmic flow (ie, papilledema).[42] Since disc swelling can be the sole manifestation,[22] anti-GQ1b antibody syndrome should be considered when the clinical presentation of patients with disc edema does not fit into that of typical optic neuritis or papilledema.

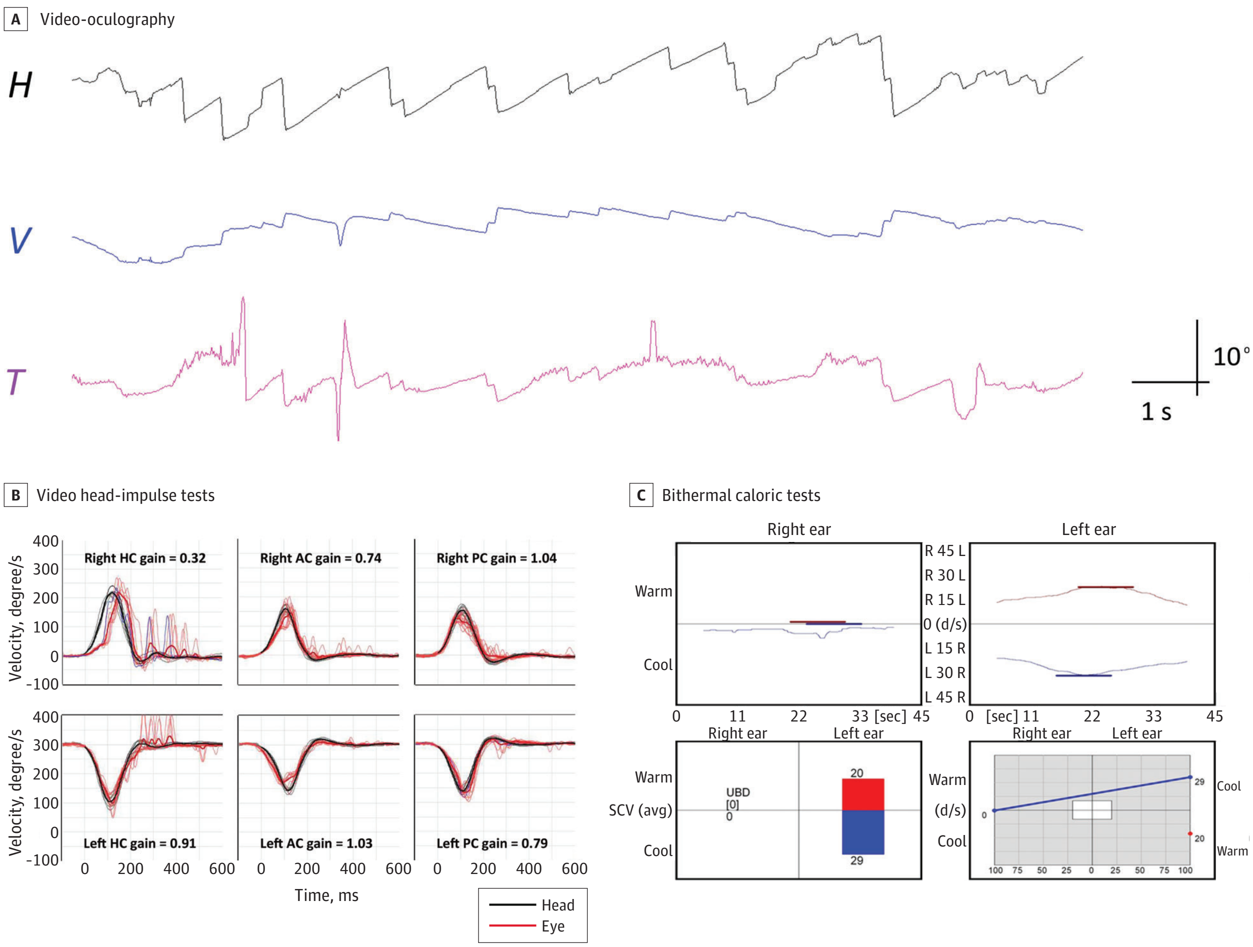
Acute Sensory Ataxic Neuropathy

The "ataxic form of polyradiculoneuritis" was first described in 1962 in a patient with profound ataxia with a loss of proprioception and vibration sensation but without weakness or ophthalmoplegia.[43] The clinical presentation of this ataxic GBS is similar to that of classic MFS except for the absence of ophthalmoplegia. Patients with ataxic GBS were found to carry anti-GQ1b immunoglobulin (Ig) G antibodies that cross-react with GT1a.[44] An autopsy revealed degeneration of the fibers in the posterior and Clarke columns but no pathology in the cerebellum.[43] This suggests that cerebellar-like ataxia is a consequence of damage of afferent fibers to the spinocerebellar system, although neither nerves nor dorsal root ganglia were sampled in the autopsy study.[9] The anti-GQ1b antibody can bind to the sensory neurons of the spinocerebellar tract and present cerebellar-like ataxia.[45] Indeed, ataxic GBS and acute sensory ataxic neuropathy may be parts of the same clinical spectrum and thus could be referred to as acute ataxic neuropathy without ophthalmoplegia.[9]

Patterns of Internal and External Ophthalmoplegia

Patients with ophthalmoplegia due to positive anti-GQ1b antibodies usually show binocular involvements but often asymmetric and monocular involvement in 27% of cases.[14] Abduction deficits are common in ophthalmoplegia (Figure 1),[14,46,47] although complete ophthalmoplegia is observed in more than half of patients.[46] Ap-

Figure 3. Patient With Acute Vestibular Syndrome



A, Video-oculography documents spontaneous nystagmus beating leftward (8.0° per second), upward (2.1° per second), and counter-clockwise (4.3° per second) without visual fixation. B, Video head-impulse tests show decreased vestibulo-ocular reflex gains and catch-up saccades for the right horizontal canal (HC) and anterior canal (AC). C, Bithermal caloric tests show a canal paresis in the right ear. H indicates horizontal position of the right eye; PC, posterior canal; SCV, slow-component velocity; T, torsional position of the right eye; UBD, unexpected beat direction; V, vertical position of the right eye.

proximately 6% of single ocular motor nerve palsy can be ascribed to anti-GQ1b antibody.[47] Additionally, patients can show internuclear ophthalmoplegia (**Video**),[48] one-and-a-half syndrome,[49] or dissociated horizontal nystagmus suggesting central involvement.[50] Notably, the most striking feature of anti-GQ1b antibody syndrome is progression of ophthalmoplegia during the illness.

Iridoplegia

Pupillary abnormalities may occur in 42% to 55% of patients with anti-GQ1b antibody syndrome, including mydriasis (42%), decreased light reflexes (42%), and anisocoria.[14,18] Given the occasional findings of light-near dissociation and denervation supersensitivity to cholinergic agents, the ciliary ganglion or short ciliary nerve may also be affected by anti-GQ1b antibodies.[51,52]

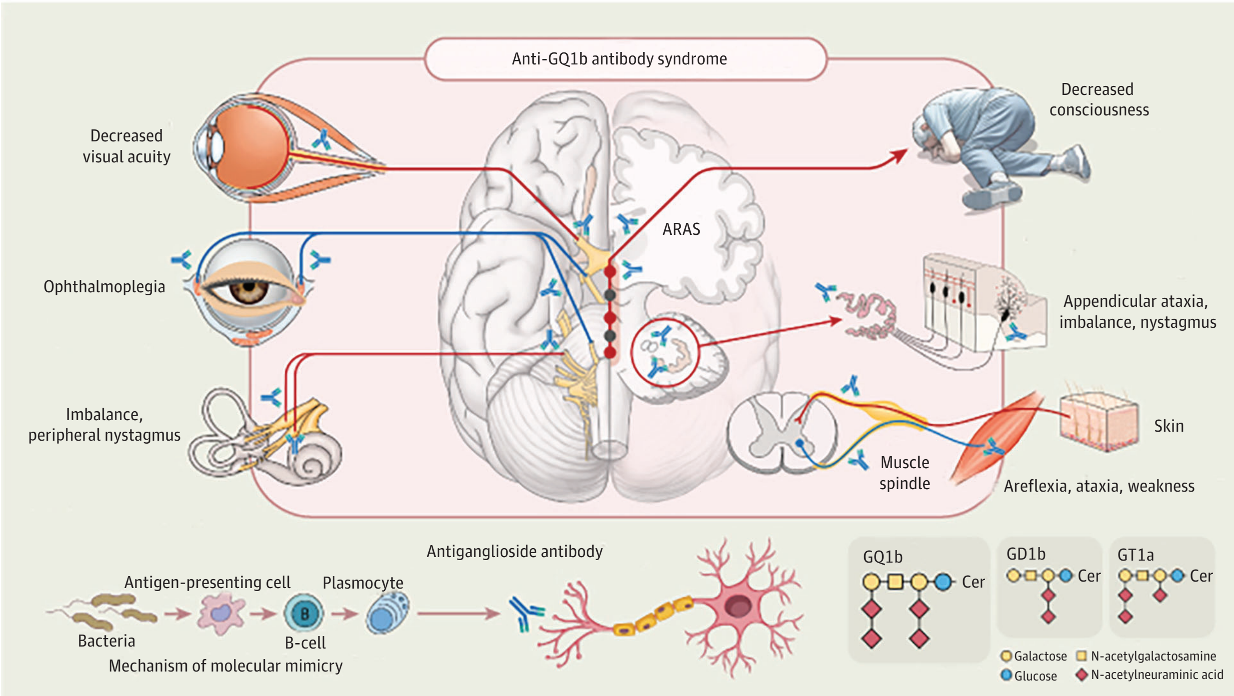
Strabismus

Diplopia and strabismus observed in patients with anti-GQ1b antibody syndrome mostly change according to gaze direction (ie, incomitant). However, the amount of ocular misalignment may be unchanged regardless of the direction of gaze or of which eye is fixating the target (ie, comitant).[53,54] In comitant strabismus, the saccadic velocity is mostly preserved.[53,54] Given the accompanying central ocular motor signs, such as spontaneous or positional downbeat nystagmus, saccadic hypermetria, or pulse-step mismatch, comitant strabismus observed in patients with anti-GQ1b antibody syndrome may be ascribed to dysfunction of the cerebellum or brainstem.[54] Similar to typical paralytic ocular motor nerve palsies, the prognosis is excellent with a full recovery mostly within 6 months.[54]

Peripheral Lesions

GQ1b gangliosides are enriched in the paranodal regions of the human oculomotor, trochlear, and abducens nerves and probably in the ciliary ganglia.[29] Thus, the clinical presentation of anti-GQ1b antibody syndrome is mostly caused by injuries to the peripheral nerves. However, GQ1b gangliosides are also found in the cerebellar granular cell layer in mice and rats.[55,56] Given the selective stain of the deep cerebellar nuclei and cerebellar molecular layer with

Figure 4. Illustration of the Clinical Findings and Mechanism of Molecular Mimicry Causing Anti-GQ1b Antibody Syndrome



ARAS indicates ascending reticular activating system; Cer, ceramide.

anti-GQ1b antibodies in humans,[17,57] these antibodies may affect the central nervous system initially or during the course when the blood-nerve barrier is broken near the roots of the cranial nerves.[58] Antiganglioside antibodies also play a role in damaging the blood-nerve barrier, allowing infiltration of inflammatory cells from the periphery and resulting in further nerve injury by destroying the tight junction between the endothelial cells forming the blood-nerve barrier.[59,60] Indeed, an autopsy revealed chromatolytic changes and pyknosis of the midbrain as well as segmental demyelination of the oculomotor and peripheral nerves, in a patient with classic MFS.[61]

Supranuclear ophthalmoplegia can also be observed in anti-GQ1b antibody syndrome. These include horizontal gaze-evoked nystagmus, preservation of the Bell phenomenon in vertical ophthalmoplegia, internuclear ophthalmoplegia, and preserved convergence in the presence of adduction palsy.[18] Moreover, patients may show diffuse slow activities on electroencephalography, absent cortical N20 in the presence of normal cervical N13 somatosensory evoked potentials, absent R2 responses on blink reflex, or elevated metabolism in the brainstem and cerebellum, which all indicate involvements of the central as well as peripheral nervous systems in anti-GQ1b antibody syndrome.[58,62]

Laboratory Findings

The anti-GQ1b antibody titers peak at clinical presentation and decline rapidly during the course of recovery. Most patients show positivity during the first week after symptom onset.[63] The anti-GQ1b IgG antibody titer correlates with clinical severity score, especially that of ophthalmoplegia.[17]

Enzyme-Linked Immunosorbent Assay vs Glycoarray

Different methods may preferentially detect different types of antibodies owing to variations in the orientation of the glycan headgroups on the immobilized surface. Thus, the results may not be concordant among different assay techniques.[64] Enzyme-linked immunosorbent assay (ELISA) remains the most commonly used method because laboratories are widely conversant with this standard technology. Glycoarrays are useful for screening different antiganglioside antibodies in a small amount of serum.[65]

The sensitivity is similar between ELISA and glycoarray, each showing rates of 89% and 87% for classic MFS, 57% and 73% for GBS-O, and the same 74% and 74% for BBE, respectively.[66] Both tests are complementary; thus, the sensitivity rates may increase up to 92% for MFS, 75% for GBS-O, and 82% for BBE when combined.[66]

Frequency of False-Negative or False-Positive Results

It is widely accepted that anti-GQ1b antibody is absent in healthy individuals or disease controls.[67] Besides, the sensitivity of serum antibodies targeting GQ1b is 92% with a specificity of 97% for MFS.[68] Thus, measurements of antibodies in the CSF confer no further benefit.[68]

Antibodies are detected in 89% to 95% of patients with MFS,[17,66,69] and the positive predictive value varies across subtypes, showing lower positivity in BBE (approximately 74%) and GBS-O (approximately 57%).[24,66,70] Thus, diagnoses based only on antibody positivity can mislead detection. Similarly, in myasthenia gravis, the presence of seronegativity in patients indicates heterogeneous pathogenesis or the presence of novel autoantibodies.[66]

Table. Clinical Characteristics and Laboratory Findings of Anti-GQ1b Antibody Syndrome

Syndrome	Clinical manifestation			Cerebrospinal fluid analysis[a]	Nerve conduction study[a]
	Ataxia	Areflexia	Others		
Typical anti-GQ1b antibody syndrome with ophthalmoplegia					
Classic Fisher syndrome	Yes	Yes	No	Albuminocytologic dissociation (~76%)	Evidence of axonal neuropathy/neuronopathy with predominant sensory nerve changes (reduced SNAP and absent H reflexes)
GBS with ophthalmoplegia	No	Yes	Weakness, sensory changes	Albuminocytologic dissociation	Suggestive of neuropathy without fulfilling demyelinating or axonal neuropathy criteria, especially in the early stage (during the first week)
Bickerstaff brainstem encephalitis	No	No	Impaired consciousness, weakness, hyperreflexia	Pleocytosis or albuminocytologic dissociation (~46%)	Normal but relevant abnormal findings when overlapped with GBS or MFS
Acute ophthalmoplegia without ataxia	No	No	No	Mostly normal; pleocytosis or albuminocytologic dissociation in limited number of patients (18%)	Normal
Atypical anti-GQ1b antibody syndrome without ophthalmoplegia					
Acute vestibular syndrome	Yes	No	Findings of peripheral and central vestibulopathy	Central vestibulopathy: pleocytosis (~20%) or albuminocytologic dissociation (~50%); peripheral: unknown	Likely normal
Optic neuropathy with disc swelling	No	No	Visual disturbance, visual field defect	Unknown	Unknown
Acute sensory ataxic neuropathy	Yes	Yes	No	Albuminocytologic dissociation (~33%)	Unknown

Abbreviations: GBS, Guillain-Barré syndrome; MFS, Miller Fisher syndrome; SNAP, sensory nerve action potential.

[a] Nerve conduction studies and cerebrospinal fluid analyses can support the diagnosis but are often inconclusive in the early stages of the disease; therefore, diagnosis and treatment should not be delayed if anti-GQ1b antibody syndrome is suspected on clinical grounds.

Association of Anti-GQ1b Antibody Presence and Titer With Prognosis

For certain subtypes, the presence and titer of the anti-GQ1b antibody can predict clinical outcomes. The clinical manifestations and prognoses of BBE do not differ between those with and without antibodies.[25] In contrast, anti-GQ1b antibody positivity is a predictive factor for artificial ventilation in patients with GBS-O.[71] An in vitro study showed that anti-GQ1b antibody has an α-latrotoxin–like blockade effect on neuromuscular transmission.[72] Likewise, intraperitoneal injection of the anti-GQ1b antibody and human serum resulted in respiratory paralysis due to blockade of neuromuscular transmission in mice.[73]

Other Antiganglioside Antibodies Associated With Anti-GQ1b Antibodies

Anti-GQ1b antibodies frequently cross-react with GT1a (up to 68%), GD3, and GD1b.[66,74,75] This cross-reactivity may explain the diverse clinical manifestations observed in patients with anti-GQ1b antibodies. For instance, patients have a higher chance of motor weakness and artificial ventilation if antibodies bind not only to GQ1b-containing antigens but also to GD1b-containing antigens.[66] Additionally, cross-reaction with the antigen containing GM1 or GD1a indicates a remarkable motor weakness.[75] Since GT1a gangliosides are abundant on the glossopharyngeal and vagus nerves, patients with anti-GQ1b antibodies cross-reacting with GT1a can exhibit profound bulbar palsy.[76] Likewise, deep sensation may be impaired when the antibody cross-reacts with GD1b.[75]

Differences Between IgM and IgG Anti-GQ1b Antibodies

It is commonly agreed that the presence of anti-GQ1b IgG antibody is likely the pathogenic isotype of anti-GQ1b antibody syndrome. However, the pathogenic role of anti-GQ1b IgM antibodies remains unclear. The presence of IgM anti-GQ1b antibody, in combination with other antibodies against a disialosyl structure (ie, GD1b, GQ1b, GT1a, and GD3), is associated with acute and chronic ataxic neuropathy.[9,77] The IgG-mediated acute immunological processes driven by exposure to bacterial antigens are different from those due to IgM antibodies. The IgM antibody persists and is due to antigen-dependent clonal B-cell expansion, thereby leading to monoclonal IgM gammopathy.[9] In this context, IgM antibodies may cause chronic damage leading to axonal degeneration, although the prognosis remains to be delineated for the various subtypes of anti-GQ1b antibody syndrome.[9]

Differential Diagnosis

Anti-GQ1b antibody syndrome should be differentiated from any disorder that manifests primarily with acute or subacute focal neurological deficits, especially ophthalmoplegia. These include Wernicke encephalopathy, multiple sclerosis, neuro-Behcet disease, myasthenia gravis, pituitary apoplexy, and Tolosa-Hunt syndrome. Additionally, patients may present with unilateral ophthalmoplegia that masquerades as ocular motor nerve palsy of various etiologies.[47] Therefore, anti-GQ1b antibody syndrome should be considered in any type of ophthalmoplegia, especially when an immune-mediated mechanism is suspected, such as an antecedent infection in young patients (usually <50 years old).[47]

Treatments

The idea of immunomodulating therapy has been endorsed by its efficacy in GBS.[78] Immunomodulating therapy may neutralize the autoantibodies and inhibit complement activation in anti-GQ1b antibody syndrome.[79] Intravenous immunoglobulin (IVIg) inhibits the binding of antibodies to GQ1b, thereby preventing complement activation and subsequent pathophysiological effects in mice.[80] Clinical data have also suggested that the use of IVIg may hasten the recovery of ophthalmoplegia and ataxia.[81] In contrast, plasmapheresis does not affect the speed of recovery in patients with anti-GQ1b antibody syndrome.[82]

The adoption of immunomodulatory therapy needs to be stratified in individual patients, given the favorable outcomes even with conservative treatment in most patients with anti-GQ1b antibody syndrome.[18,81] To date, no well-designed randomized clinical trials have determined the therapeutic effects of the use of IVIg or plasmapheresis in comparison with placebo.[83]

Prognosis

Most patients recover completely within 4 to 12 weeks of symptom onset. The median duration to full recovery is approximately 4 weeks for ataxia and 12 weeks for ophthalmoplegia.[18] The recovery rate showed no correlation with age, sex, evidence of prior infection, and disability at the nadir of the patients.[18] Rarely, patients deteriorate despite treatment, developing the features of GBS and eventually requiring mechanical ventilation.[84] This neurologic deterioration occurs more frequently in BBE and GBS-O subtypes.[84,85] Thus, careful monitoring and management are required for patients with certain subtypes because serious complications can occur: eg, coma,[86] cardiomyopathy due to dysautonomia,[87] corticobulbar dysfunction,[67,88] and even death.

Potential of Anti-GQ1b Antibody Syndrome Relapse

Anti-GQ1b antibody syndrome is mostly monophasic, with approximate recurrences in only 14% of cases.[89,90] Relapse can occur, especially when the patients have an underlying disease such as HIV infection[32] or cancer, which may expose immunogens with molecular mimicry. Alternatively, given the high frequency of human histocompatibility leukocyte antigen DR2 in patients with recurrence,[89] a particular immunogenetic background may play a role. During relapse, the antibody titer is higher than that during the first attack with a grave clinical presentation.[91] The overall clinical outcome remains favorable despite multiple recurrences.[90]

Paraneoplastic Etiology

During the process of carcinogenesis, novel gangliosides may be expressed on the tumor cells that share epitopes with cranial or peripheral nerves.[92] Consequently, paraneoplastic cranial neuropathy can be associated with various types of cancers, eg, thyroid carcinoma, lung cancer, and lymphomas.[93] Elevated antiganglioside antibody levels in patients with thyroid cancer may suggest that the process of carcinogenesis can evoke autoimmunization.[94] Compared with patients without cancer, those with cancer may show a higher IgM antiganglioside antibody titer due to an antitumor immune reaction itself, irrespective of the presence of cranial neuropathy.[95] Thus, it is still unknown whether antiganglioside antibodies are detected as a chance association or produced due to an immunological mechanism triggered by the tumor.[96]

Conclusions

The clinical spectrum of anti-GQ1b antibody syndrome is expanding. However, a causal relationship between antibody positivity and certain phenotypes remains to be established. Given the various clinical features of anti-GQ1b antibody syndrome, recognition of its wide clinical spectrum may aid in the differentiation and management of immune-mediated neuropathies with multiple faces.

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